

DLOPs Assignment-1

Deadline : 24/01/2026 11:59 PM

Submission Guidelines:

1. Use Python and PyTorch for performing experiments.
 2. Submit a single report with all results and the detailed analysis.
 3. **Share a Colab link in the report that includes already run experiments. Without the Colab link, the student will get 0 Marks. Do not share code files separately.**
 4. **Report file naming convention: Rollnumber_Name_Ass1.pdf. For example: B22CS043_Firstname_Surname_Ass1.pdf**
 5. Follow all the GitHub guidelines mentioned below.
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Github Guidelines:

- Create your GitHub profile.
 - Create a repository named: MLOps-Name-rollNumber
 - Create a branch named Assignment 1.
 - In this branch, push the .ipynb file and results (**one** best model, train-val graphs, results, etc.) and report.
 - Your readme for this assignment should contain the below mentioned tables and results.
 - Create GitHub pages for your repository, which needs to be updated after each assignment.
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Q1(a). Train a deep learning model on the MNIST and FashionMNIST datasets with 70%-10%-20% train-val-test split. Use the Following Models: ResNet-18 and ResNet-50. Fill in the classification accuracy in the table below and provide a detailed analysis based on the results.

			Test Classification Accuracy (in %)	
Batch Size	Optimizers	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001		
16	SGD	0.0001		
16	Adam	0.001		
16	Adam	0.0001		
32	SGD	0.001		
32	SGD	0.0001		
32	Adam	0.001		

32	Adam	0.0001		
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Note: Also perform experiments by varying hyperparameters.

pin_memory=False, True

Epochs = vary the number of epochs by yourself (perform an experiment with any two total numbers of epochs).

Keep these factors Constant:- USE_AMP=True

The valid experiments include achieving a Classification accuracy of more than 80+%

Q1(b). Also train a SVM classifier on MNIST and FashionMNIST datasets with varying SVM hyperparameters with kernels such as: 'poly', 'rbf'. Report Testing Classification Accuracy with training time (in ms).

Q2. For the **FashionMNIST** dataset, report performance and train-time based on running the experiments on CPU and GPU both. Write a detailed report analyzing the training time, number of FLOPs used and the classification accuracy of the models.

Compu te	Batch Size	Optimi zer	Learn ing Rate	Test Classification Accuracy (in %)			Train Time (in ms)			FLOPs				
				Res Net-18	Res Net-32	Res Net-50	Res Net-18	Res Net-32	Res Net-50	ResNet-18	ResNet-32	ResNet-50		
				CPU	16	SGD	0.001							
				CPU	16	Adam	0.001							
				GPU	16	SGD	0.001							
				GPU	16	Adam	0.001							